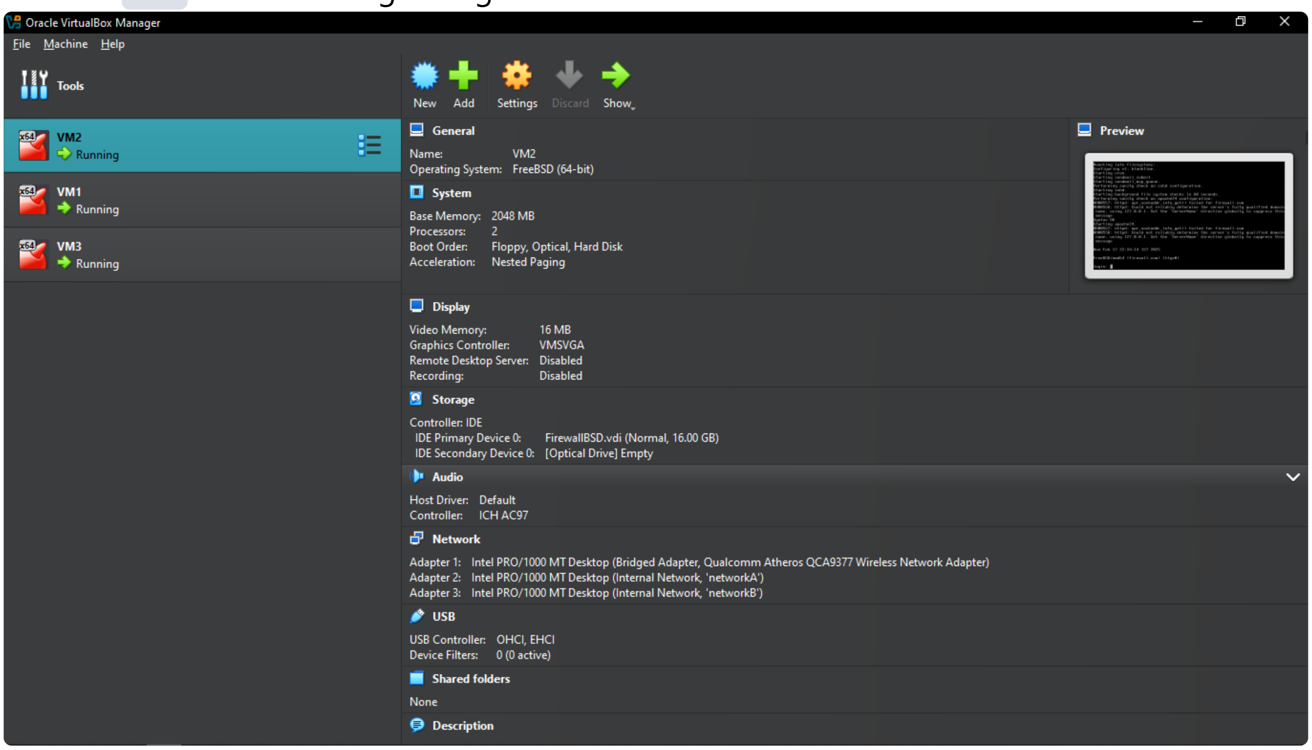
24m0847 CS790 Assignment Task 1

Task 1

Requirements:

1. One VM running FreeBSD 13.4 (preferred) with no X window. You could use a very lean configuration — e.g., 2 cores and 2 GB RAM. This should be acting as a L3 forwarding firewall. You need to enable the IP routing for the same. We would rely on FreeBSD pf firewall for this exercise.

1. Created `VM2` with following configurations:



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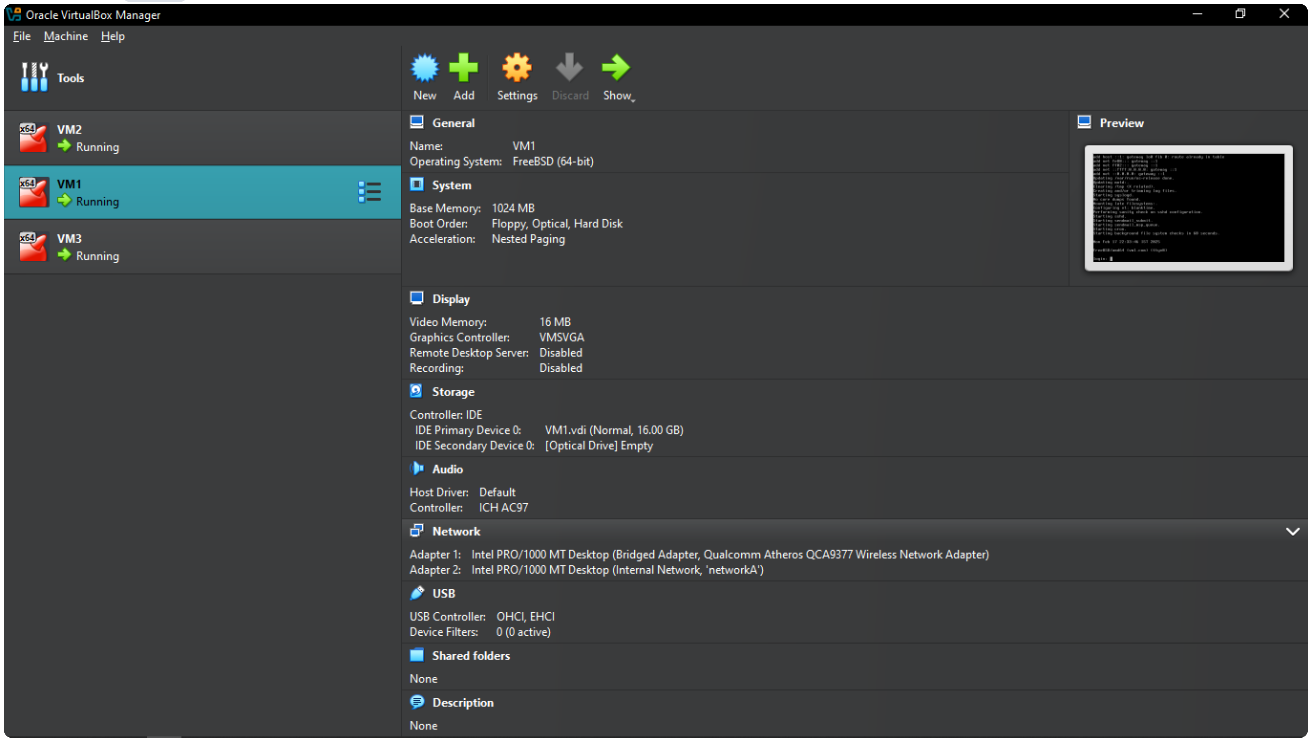
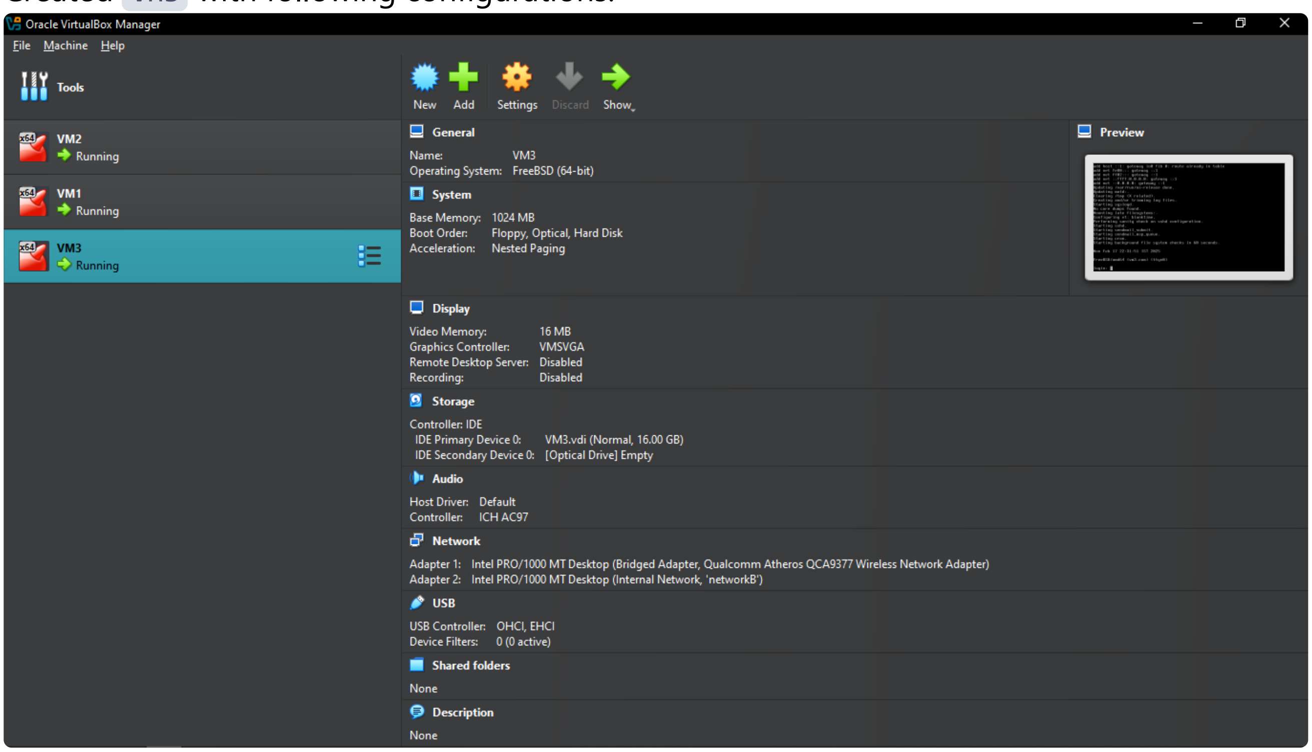
2. Two other VMs running any OS of your choice (Linux or FreeBSD) with as minimal configuration as possible.

1. Created `VM1` with following configurations:



2. Created `VM3` with following configurations:



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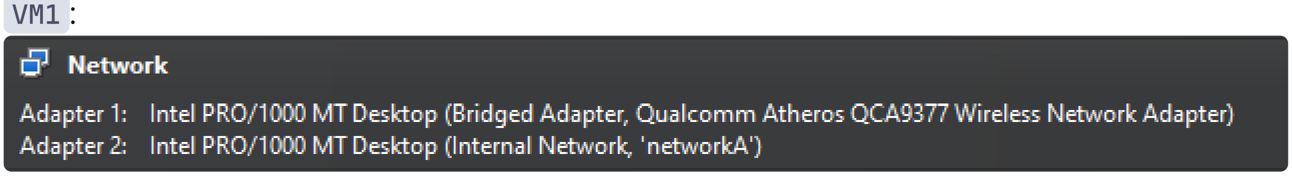
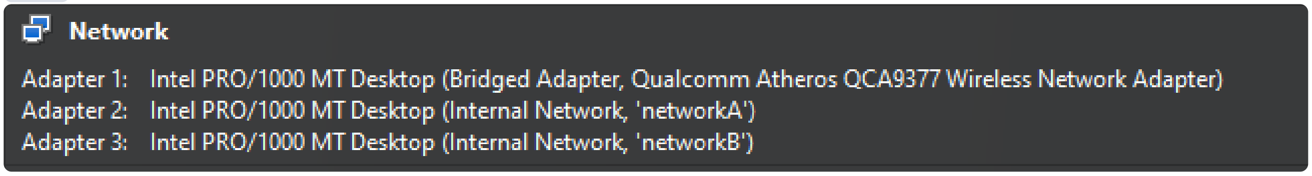
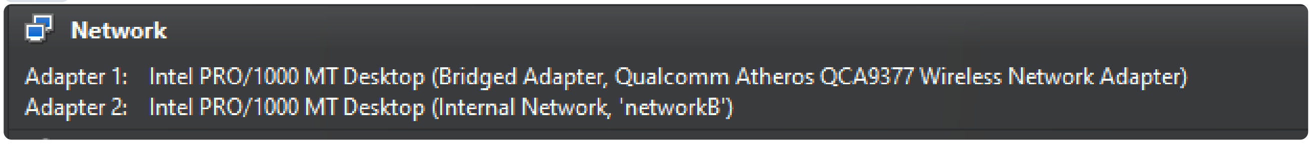
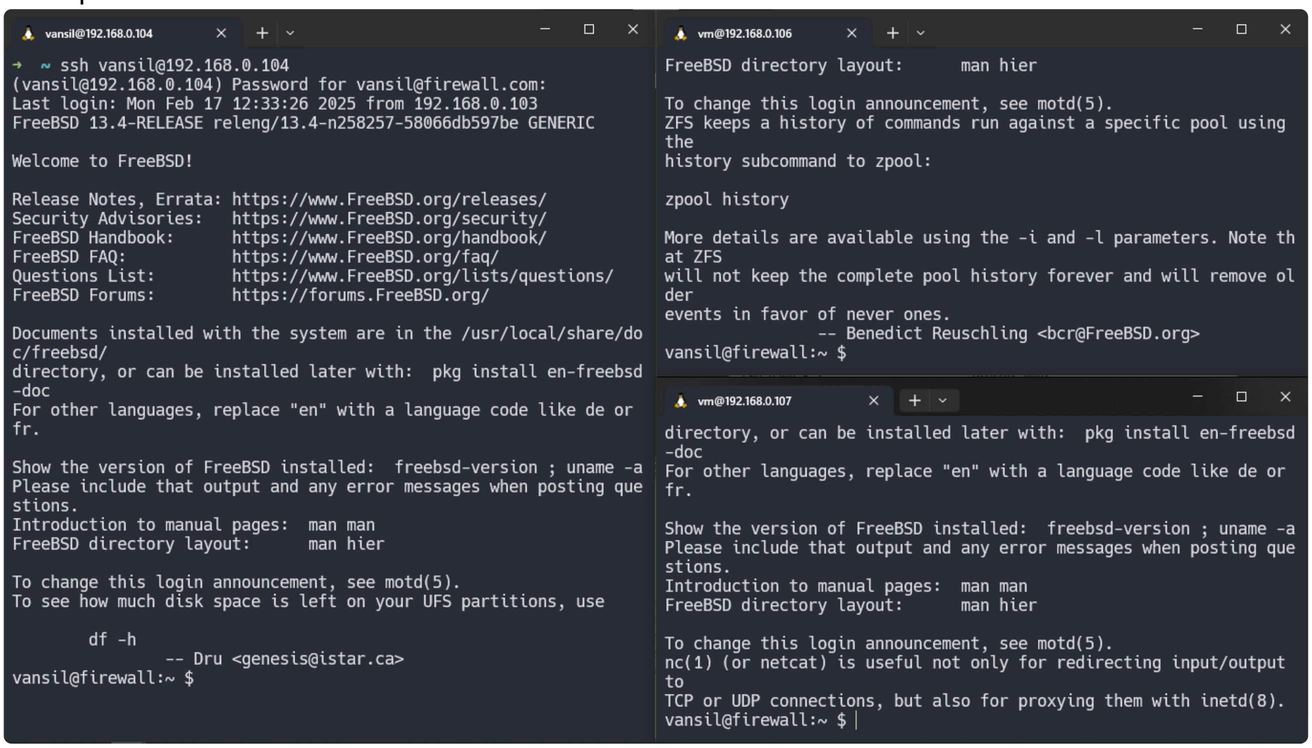
3. The three VMs must be connected using virtual networks.

- `VM1`:



- `VM2`:


- `VM3`:


- `Adapter1` is **Bridged Adapter** for doing `SSH` from `Host OS` to access all 3 `Guest OSes`.
- Example:


- We will look at this in details in upcoming answers.

Objective: The The assignment is designed to allow you to get your hands dirty with FreeBSD pf (an industry standard firewall framework) to filter traffic.

- You would require configuring the three VMs using the configurational setup

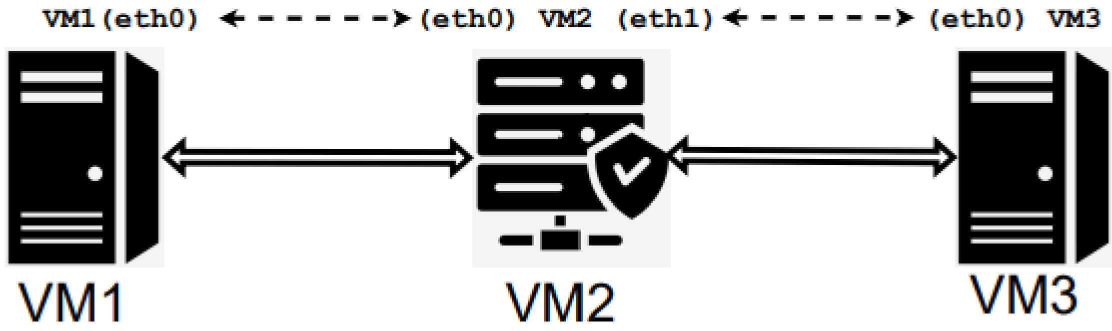


Figure 1: VM network arrangement.

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a) Assign private IP addresses to each interface so the VMs can ping one another. VM1 (eth0) and VM2 (eth0) should be in the same subnet. VM2(eth1) and VM3 (eth0) should be in the same subnet. (5 points)

- Created 2 Subnets:

$$networkA = 192.168.25.0/30$$

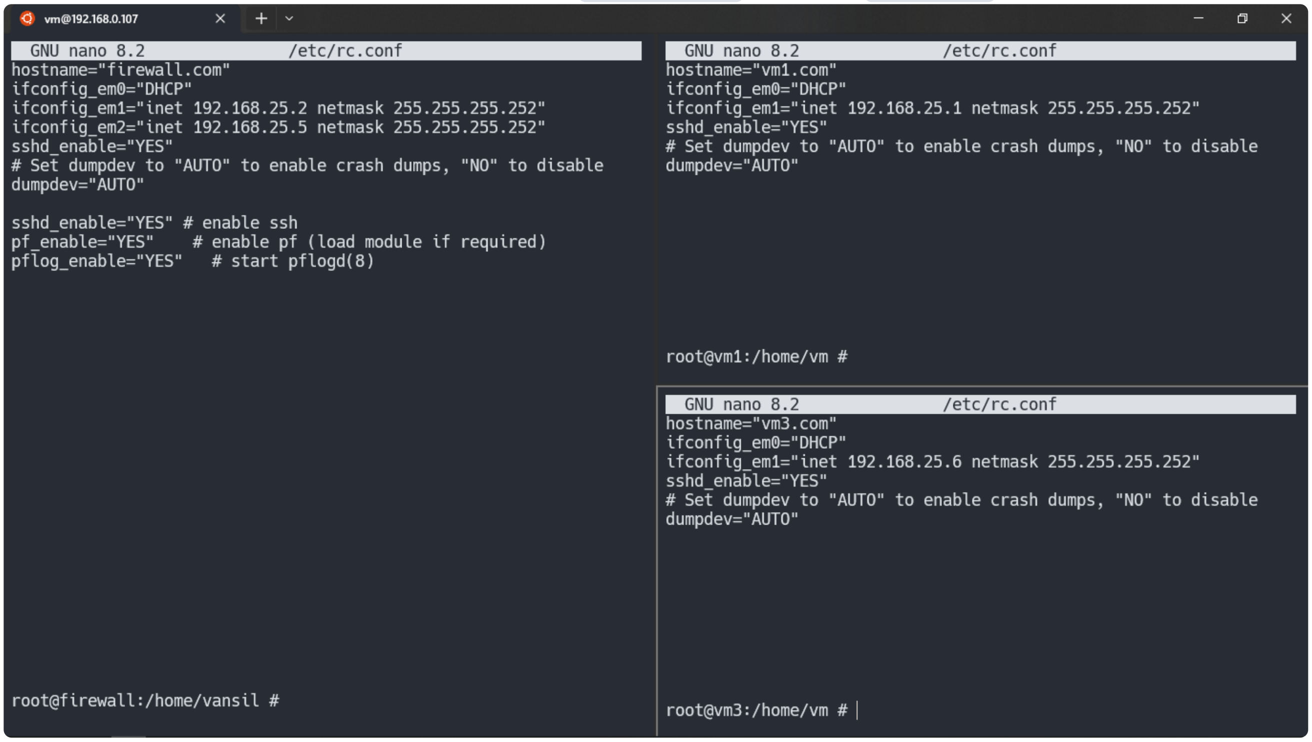
$$networkB = 192.168.25.4/30$$

- Private IP addresses to each interface:

```
# VM1
em0 : 192.168.0.106  # used for SSH from Host OS (DHCP)
em1 : 192.168.25.1  # networkA (Static IP)

# VM2
em0 : 192.168.0.104 # used for SSH from Host OS (DHCP)
em1 : 192.168.25.2 # networkA (Static IP)
em2 : 192.168.25.5 # networkB (Static IP)

# VM3
em0 : 192.168.0.107 # used for SSH from Host OS (DHCP)
em1 : 192.168.25.6 # networkB (Static IP)
```

- How to update IP addresses?
  - edit `/etc/rc.conf` as following...

- machine ips (192.168.25.x) setup (ignore `PF_enable` lines as `pf.conf` was empty)



```
GNU nano 8.2                /etc/rc.conf
hostname="firewall.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.2 netmask 255.255.255.252"
ifconfig_em2="inet 192.168.25.5 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

sshd_enable="YES" # enable ssh
pf_enable="YES"    # enable pf (load module if required)
pflog_enable="YES"   # start pflogd(8)

root@firewall:/home/vansil #
```

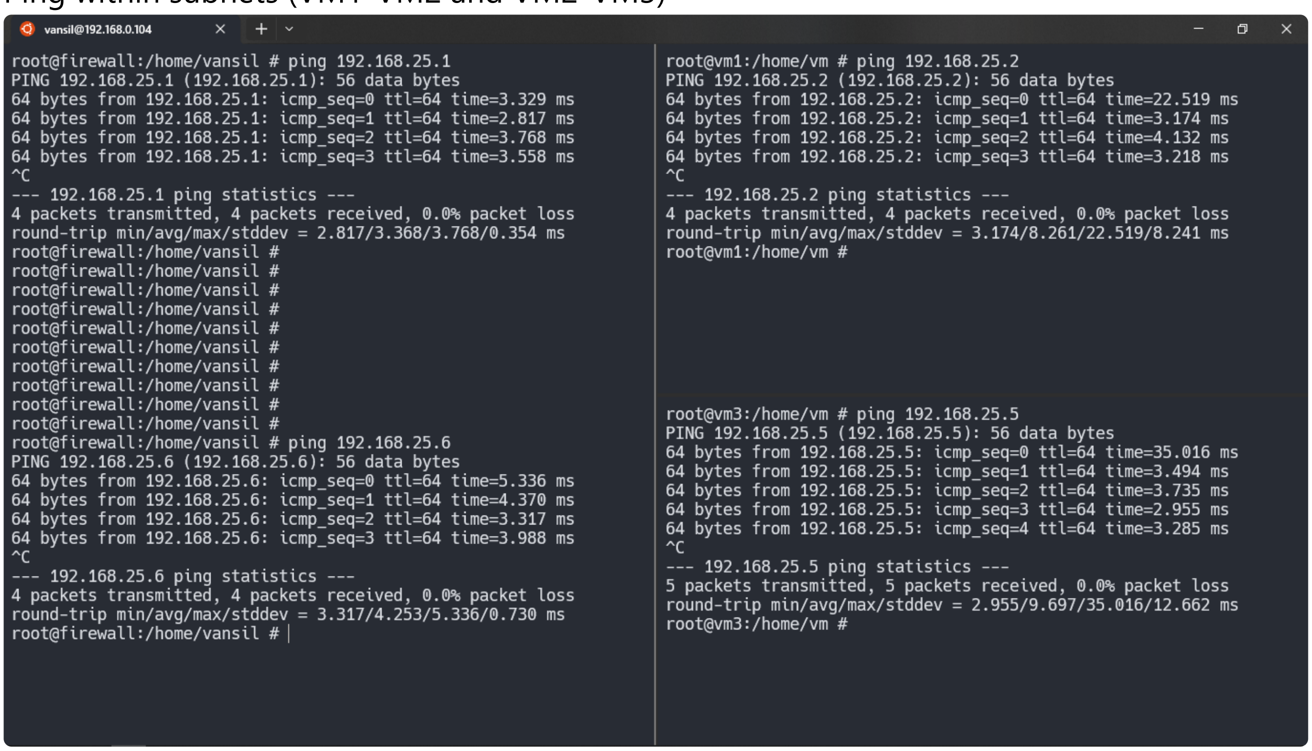
```
GNU nano 8.2                /etc/rc.conf
hostname="vm1.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.1 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

root@vm1:/home/vm #
```

```
GNU nano 8.2                /etc/rc.conf
hostname="vm3.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.6 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

root@vm3:/home/vm #
```

- Reset network interfaces

```
/etc/rc.d/netif restart && /etc/rc.d/routing restart
```

- Ping within subnets (VM1-VM2 and VM2-VM3)



```
root@firewall:/home/vansil # ping 192.168.25.1
PING 192.168.25.1 (192.168.25.1): 56 data bytes
64 bytes from 192.168.25.1: icmp_seq=0 ttl=64 time=3.329 ms
64 bytes from 192.168.25.1: icmp_seq=1 ttl=64 time=2.817 ms
64 bytes from 192.168.25.1: icmp_seq=2 ttl=64 time=3.768 ms
64 bytes from 192.168.25.1: icmp_seq=3 ttl=64 time=3.558 ms
^C
--- 192.168.25.1 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.817/3.368/3.768/0.354 ms
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil #
root@firewall:/home/vansil # ping 192.168.25.6
PING 192.168.25.6 (192.168.25.6): 56 data bytes
64 bytes from 192.168.25.6: icmp_seq=0 ttl=64 time=5.336 ms
64 bytes from 192.168.25.6: icmp_seq=1 ttl=64 time=4.370 ms
64 bytes from 192.168.25.6: icmp_seq=2 ttl=64 time=3.317 ms
64 bytes from 192.168.25.6: icmp_seq=3 ttl=64 time=3.988 ms
^C
--- 192.168.25.6 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.317/4.253/5.336/0.730 ms
root@firewall:/home/vansil #
```

```
root@vm1:/home/vm # ping 192.168.25.2
PING 192.168.25.2 (192.168.25.2): 56 data bytes
64 bytes from 192.168.25.2: icmp_seq=0 ttl=64 time=22.519 ms
64 bytes from 192.168.25.2: icmp_seq=1 ttl=64 time=3.174 ms
64 bytes from 192.168.25.2: icmp_seq=2 ttl=64 time=4.132 ms
64 bytes from 192.168.25.2: icmp_seq=3 ttl=64 time=3.218 ms
^C
--- 192.168.25.2 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.174/8.261/22.519/8.241 ms
root@vm1:/home/vm #
```

```
root@vm3:/home/vm # ping 192.168.25.5
PING 192.168.25.5 (192.168.25.5): 56 data bytes
64 bytes from 192.168.25.5: icmp_seq=0 ttl=64 time=35.016 ms
64 bytes from 192.168.25.5: icmp_seq=1 ttl=64 time=3.494 ms
64 bytes from 192.168.25.5: icmp_seq=2 ttl=64 time=3.735 ms
64 bytes from 192.168.25.5: icmp_seq=3 ttl=64 time=2.955 ms
64 bytes from 192.168.25.5: icmp_seq=4 ttl=64 time=3.285 ms
^C
--- 192.168.25.5 ping statistics ---
5 packets transmitted, 5 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.955/9.697/35.016/12.662 ms
root@vm3:/home/vm #
```

- Within subnet ping is working but `IP forwarding` is not working at this point. For that, let's move to next requirement `1.1.b`.

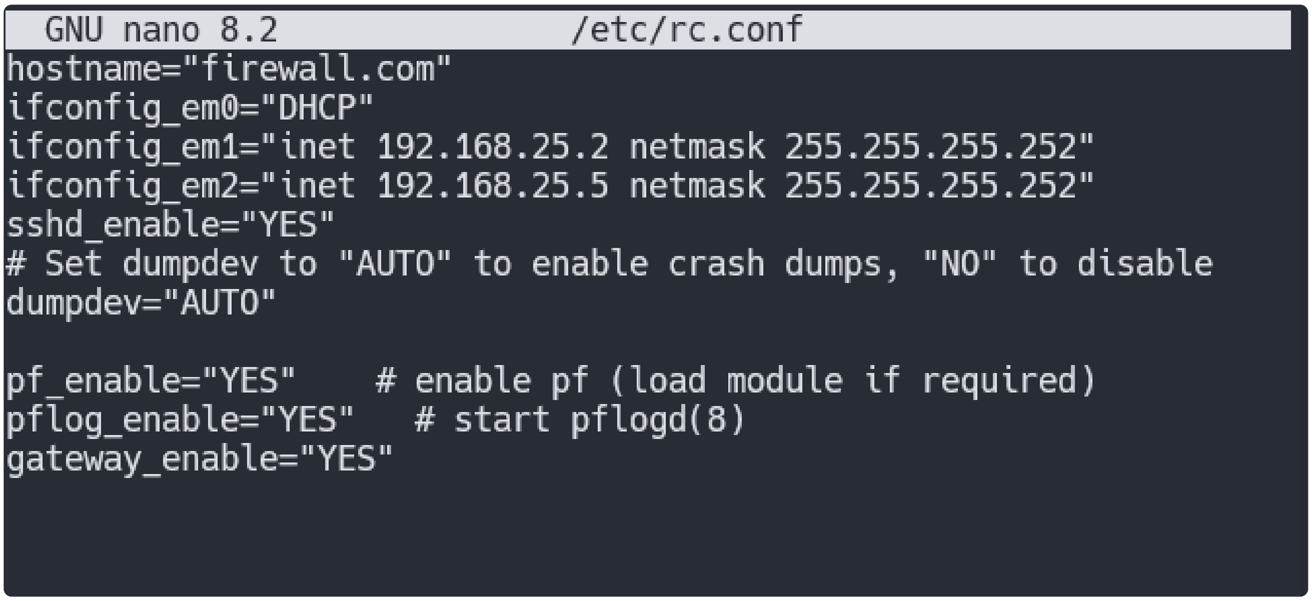
---

b) Setup IP forwarding on VM2 such that VM1 can ping both interfaces IPs of VM2 and VM3. Make sure traffic from VM1 to VM3 goes via VM2 (can be seen via traceroute). (5 points)

- Enable IP Forwarding on `vm2`:
- Enable gateway (`/etc/rc.conf`):

```
gateway_enable="YES"
```

- VM2



```
  GNU nano 8.2                    /etc/rc.conf
hostname="firewall.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.2 netmask 255.255.255.252"
ifconfig_em2="inet 192.168.25.5 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

pf_enable="YES"    # enable pf (load module if required)
pflog_enable="YES"   # start pflogd(8)
gateway_enable="YES"
```

- restart service routing

```
service routing restart
```

- VM2

```
root@firewall:/home/vansil # service routing restart
delete host 127.0.0.1: gateway lo0
delete host ::1: gateway lo0
delete net fe80::: gateway ::1
delete net ff02::: gateway ::1
delete net ::ffff:0.0.0.0: gateway ::1
delete net ::0.0.0.0: gateway ::1
add host 127.0.0.1: gateway lo0
Additional inet routing options: gateway=YES.
add host ::1: gateway lo0
add net fe80::: gateway ::1
add net ff02::: gateway ::1
add net ::ffff:0.0.0.0: gateway ::1
add net ::0.0.0.0: gateway ::1
root@firewall:/home/vansil #
```

- Set routes on VM1 and VM3
- VM1 (Route to VM3 via VM2)

```
static_routes="networkB"
route_networkB="-net 192.168.25.4/30 192.168.25.2"
```

- VM3 (Route to VM1 via VM2)

```
static_routes="networkA"
route_networkA="-net 192.168.25.0/30 192.168.25.5"
```

- `/etc/rc.conf`

```
vm@192.168.0.106
 GNU nano 8.2                    /etc/rc.conf
hostname="vm1.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.1 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"
static_routes="networkB"
route_networkB="-net 192.168.25.4/30 192.168.25.2"

root@vm1:/home/vm #
```

```
vm@192.168.0.107
 GNU nano 8.2                    /etc/rc.conf
hostname="vm3.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.6 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"
static_routes="networkA"
route_networkA="-net 192.168.25.0/30 192.168.25.5"

root@vm3:/home/vm #
```

- Add routes to VM3

```
root@vm3:/home/vm # route add -net 192.168.25.0/30 192.168.25.5
add net 192.168.25.0: gateway 192.168.25.5
root@vm3:/home/vm #
```

- Add routes to VM1

```
root@vm1:/home/vm # route add -net 192.168.25.4/30 192.168.25.2
add net 192.168.25.4: gateway 192.168.25.2
root@vm1:/home/vm #
```

- Check routing on VM2

```
netstat -rn
```

- netstat

```
root@firewall:/home/vansil # netstat -rn | grep 192.168.25
192.168.25.0/30    link#2            U           em1
192.168.25.2       link#2            UHS         lo0
192.168.25.4/30    link#3            U           em2
192.168.25.5       link#3            UHS         lo0
root@firewall:/home/vansil #
```

- ping from `VM1 to VM3` and `VM3 to VM1`

vm@192.168.0.106

```
--- 192.168.25.6 ping statistics ---
3 packets transmitted, 0 packets received, 100.0% packet loss
root@vm1:/home/vm # route add -net 192.168.25.4/30 192.168.25.2
add net 192.168.25.4: gateway 192.168.25.2
root@vm1:/home/vm # ping 192.168.25.6
PING 192.168.25.6 (192.168.25.6): 56 data bytes
64 bytes from 192.168.25.6: icmp_seq=0 ttl=63 time=13.788 ms
64 bytes from 192.168.25.6: icmp_seq=1 ttl=63 time=5.976 ms
64 bytes from 192.168.25.6: icmp_seq=2 ttl=63 time=6.633 ms
64 bytes from 192.168.25.6: icmp_seq=3 ttl=63 time=6.723 ms
^C
--- 192.168.25.6 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 5.976/8.280/13.788/3.193 ms
root@vm1:/home/vm #
```

vm@192.168.0.107

```
To quickly create an empty file, use "touch filename".
                -- Dru <genesis@istar.ca>
vm@vm3:~ $ su
Password:
root@vm3:/home/vm # route add -net 192.168.25.0/30 192.168.25.5
add net 192.168.25.0: gateway 192.168.25.5
root@vm3:/home/vm # ping 192.168.25.1
PING 192.168.25.1 (192.168.25.1): 56 data bytes
64 bytes from 192.168.25.1: icmp_seq=0 ttl=63 time=5.767 ms
64 bytes from 192.168.25.1: icmp_seq=1 ttl=63 time=6.276 ms
64 bytes from 192.168.25.1: icmp_seq=2 ttl=63 time=6.264 ms
^C
--- 192.168.25.1 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 5.767/6.102/6.276/0.237 ms
root@vm3:/home/vm #
```

- traceroute from `VM1 to VM3`

```
root@vm1:/home/vm # traceroute 192.168.25.6
traceroute to 192.168.25.6 (192.168.25.6), 64 hops max, 40 byte pa
ckets
 1  192.168.25.2 (192.168.25.2)  3.293 ms  2.981 ms  4.203 ms
 2  192.168.25.6 (192.168.25.6)  6.316 ms  6.354 ms  6.133 ms
root@vm1:/home/vm #
```

- traceroute from `VM3 to VM1`

```
root@vm3:/home/vm # traceroute 192.168.25.1
traceroute to 192.168.25.1 (192.168.25.1), 64 hops max, 40 byte pa
ckets
 1  192.168.25.5 (192.168.25.5)  3.310 ms  6.957 ms  4.959 ms
 2  192.168.25.1 (192.168.25.1)  7.034 ms  5.960 ms  7.805 ms
root@vm3:/home/vm # |
```

- Working Finally!

---

c) On VM3, install an HTTP server (e.g., Apache) on the VM3. Create a web directory and add some files to it. Enable the webserver process on port 80. (2 points)

- Install Apache

```
pkg install apache24
sysrc apache24_enable="YES"
service apache24 start
```

- Not able to use internet.
- Default Gateway issue.
- Add default gateway using `route add default 192.168.0.1` (em0) to hostel wifi router
- VM2:

```
root@firewall:/home/vansil # route add default 192.168.0.1
add net default: gateway 192.168.0.1
```

- Apache installing using `pkg install apache24` on `VM2`

```
=====
Message from apache24-2.4.62:

--
To run apache www server from startup, add apache24_enable="yes"
in your /etc/rc.conf. Extra options can be found in startup script
.

Your hostname must be resolvable using at least 1 mechanism in
/etc/nsswitch.conf typically DNS or /etc/hosts or apache might
have issues starting depending on the modules you are using.


- apache24 default build changed from static MPM to modular MPM
- more modules are now enabled per default in the port
- icons and error pages moved from WWWDIR to DATADIR

   If build with modular MPM and no MPM is activated in
   httpd.conf, then mpm_prefork will be activated as default
   MPM in etc/apache24/modules.d to keep compatibility with
   existing php/perl/python modules!

Please compare the existing httpd.conf with httpd.conf.sample
and merge missing modules/instructions into httpd.conf!
```

- Enable Apache service at startup

```
sysrc apache24_enable="YES"
service apache24 start
```

- VM2

```
root@firewall:/home/vansil # sysrc apache24_enable="YES"
apache24_enable:  -> YES
root@firewall:/home/vansil # service apache24 start
Performing sanity check on apache24 configuration:
Syntax OK
Starting apache24.
root@firewall:/home/vansil #
```

- Verify Apache is running

```
service apache24 status
```

- VM2

```
root@firewall:/home/vansil # service apache24 status
apache24 is running as pid 3443.
root@firewall:/home/vansil # 
```

- Update default gateway for `VM1` and `VM3`:

vm@192.168.0.106

```
192.168.25.1       link#2             UHS          lo0
192.168.25.4/30    192.168.25.2       UGS          em1
root@vm1:/home/vm # pkg update
Updating FreeBSD repository catalogue...
pkg: An error occured while fetching package
pkg: An error occured while fetching package
repository FreeBSD has no meta file, using default settings
pkg: An error occured while fetching package
pkg: An error occured while fetching package
pkg: An error occured while fetching package
pkg: An error occured while fetching package
Unable to update repository FreeBSD
Error updating repositories!
root@vm1:/home/vm # route add default 192.168.0.1
add net default: gateway 192.168.0.1
root@vm1:/home/vm #
```

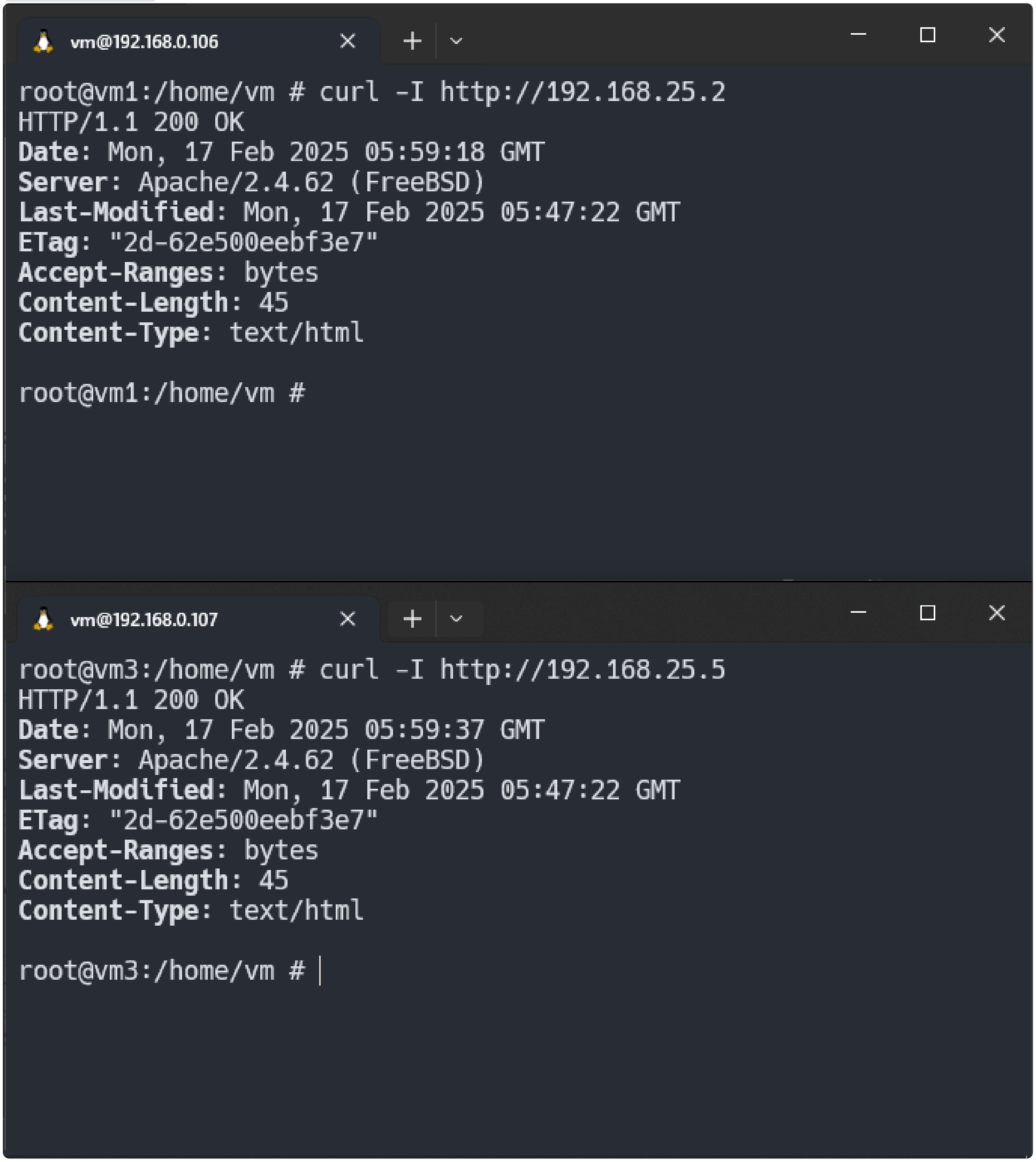
vm@192.168.0.107

```
root@vm3:/home/vm # netstat -rn | grep 192.
192.168.0.0/24     link#1             U            em0
192.168.0.107      link#1             UHS          lo0
192.168.25.0/30    192.168.25.5       UGS          em1
192.168.25.4/30    link#2             U            em1
192.168.25.6       link#2             UHS          lo0
root@vm3:/home/vm # route 192.168.0.1
route: bad keyword: 192.168.0.1
route: usage: route [-j jail] [-46dnqtv] command [[modifiers] args
]
root@vm3:/home/vm # route add 192.168.0.1
route: writing to routing socket: Invalid argument
add host 192.168.0.1 fib 0: Invalid argument
root@vm3:/home/vm # route add default 192.168.0.1
add net default: gateway 192.168.0.1
root@vm3:/home/vm # 
```

- Install curl using `pkg install curl` on `VM1` and `VM3`
- Testing using curl from `VM1` and `VM2`

```
curl -I http://192.168.25.2
curl -I http://192.168.25.5
```

- `VM1 and VM3`:



- Create a web directory and add some files.

- Apache's default web root `/usr/local/www/apache24/data`

```
root@firewall:/home/vansil # cd /usr/local/www/apache24
root@firewall:/usr/local/www/apache24 # ls
cgi-bin data    error   icons
root@firewall:/usr/local/www/apache24 # cd data
root@firewall:/usr/local/www/apache24/data # ls
index.html
root@firewall:/usr/local/www/apache24/data #
```

- create `web` directory:

```
mkdir -p web
```

- VM2:

```
root@firewall:/usr/local/www/apache24/data # mkdir -p web
root@firewall:/usr/local/www/apache24/data # ls
index.html      web
root@firewall:/usr/local/www/apache24/data # tree
.
├── index.html
└── web

2 directories, 1 file
root@firewall:/usr/local/www/apache24/data #
```

- Add files `index.html`

```
<!-- index.html -->
This is an Apache server running on VM2
```

- Set correct ownership and permissions: Ownership of Directory

```
chown -R www:www /usr/local/www/apache24/web
chmod -R 755 /usr/local/www/apache24/web
```

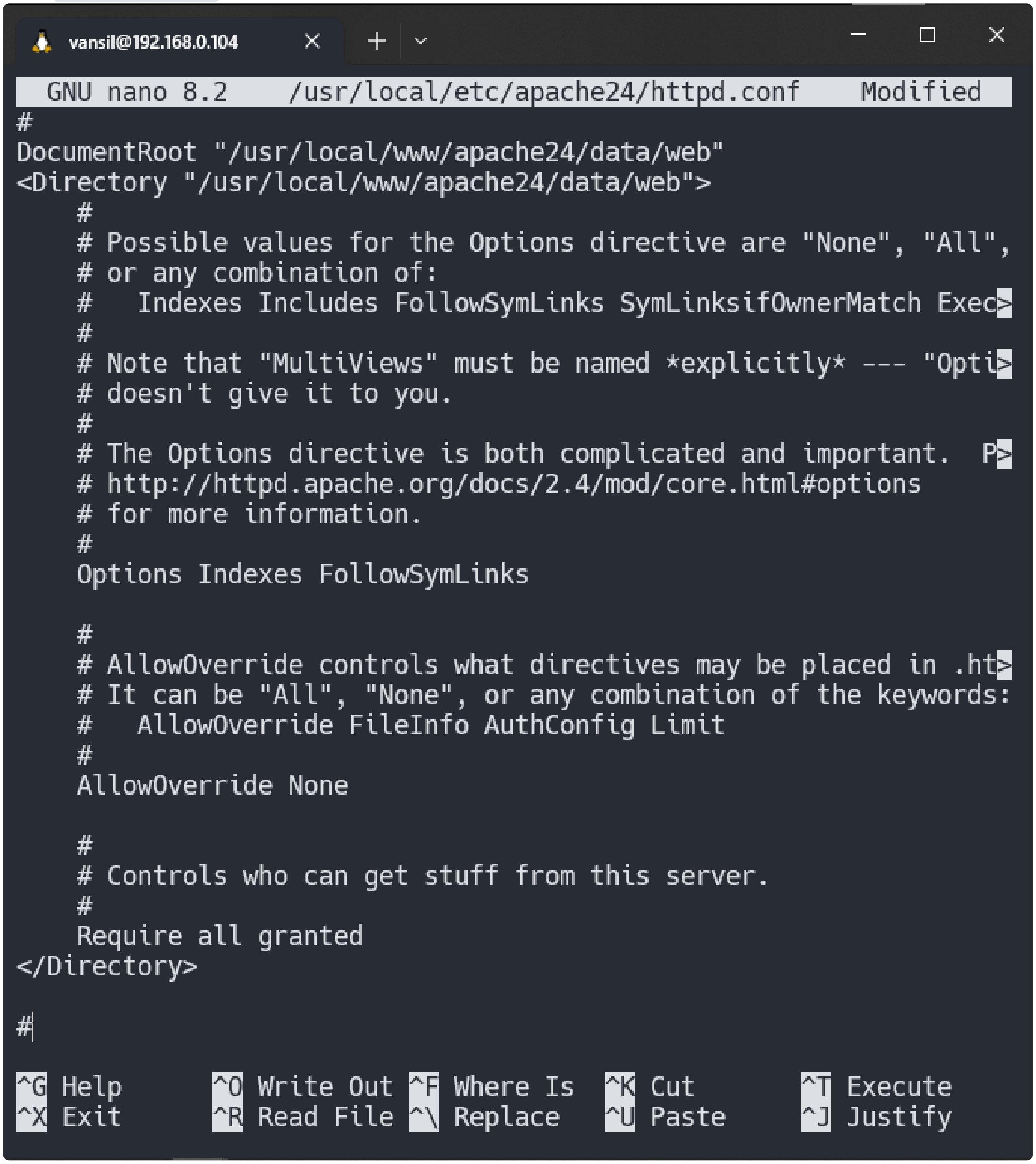
- for `web` directory

```
root@firewall:/usr/local/www/apache24/data # chown -R www:www web
root@firewall:/usr/local/www/apache24/data # chmod -R 755 web
root@firewall:/usr/local/www/apache24/data #
```

- Configure Apache to use new directory

```
nano /usr/local/etc/apache24/http.conf
```

- Edit `DocumentRoot`



- Restart `apache24`

```
root@firewall:/usr/local/www/apache24/data # service apache24 rest
art
Performing sanity check on apache24 configuration:
Syntax OK
Stopping apache24.
Waiting for PIDS: 3443.
Performing sanity check on apache24 configuration:
Syntax OK
Starting apache24.
root@firewall:/usr/local/www/apache24/data # 
```

- Check using command `curl http://192.168.25.x/index.html`
- Check in `VM1`

```
root@vm1:/home/vm # curl http://192.168.25.2/index.html
This is an Apache server running on VM2
root@vm1:/home/vm # 
```

- Check in `VM3`

```
root@vm3:/home/vm # curl http://192.168.25.5/index.html
This is an Apache server running on VM2
root@vm3:/home/vm # 
```

- Run Apache server on port 80:
- Check on which port it is currently running:

```
grep -i "listen" /usr/local/etc/apache24/httpd.conf
```

- Running on port `80` by default

```
root@firewall:/usr/local/www/apache24/data # grep -i "listen" /usr
/local/etc/apache24/httpd.conf
# Listen: Allows you to bind Apache to specific IP addresses and/o
r
# Change this to Listen on specific IP addresses as shown below to

#Listen 12.34.56.78:80
Listen 80
root@firewall:/usr/local/www/apache24/data # 
```

---

d) You need to use pf on VM2, to restrict access only to port 80 and not allow anything else. To test the correct functionality, you could install an SSH server on VM3 and try to access it. Correct firewall functionality should restrict access to port 22 (SSH). (13 points)

- Check PF status

```
pfctl -s info
```

- VM2

```
root@firewall:~ # pfctl -s info
Status: Enabled for 0 days 01:50:27           Debug: Urgent

State Table                          Total             Rate
  current entries                        0
  searches                           40470            6.1/s
  inserts                                0            0.0/s
  removals                               0            0.0/s
Counters
  match                              40470            6.1/s
  bad-offset                             0            0.0/s
  fragment                               0            0.0/s
  short                                  0            0.0/s
  normalize                              0            0.0/s
  memory                                 0            0.0/s
  bad-timestamp                          0            0.0/s
  congestion                             0            0.0/s
  ip-option                             52            0.0/s
  proto-cksum                            0            0.0/s
  state-mismatch                         0            0.0/s
  state-insert                           0            0.0/s
  state-limit                            0            0.0/s
  src-limit                              0            0.0/s
  synproxy                               0            0.0/s
  map-failed                             0            0.0/s
root@firewall:~ #
```

- `/etc/rc.conf`

```
root@firewall:/etc # cat /etc/rc.conf
hostname="firewall.com"
ifconfig_em0="DHCP"
ifconfig_em1="inet 192.168.25.2 netmask 255.255.255.252"
ifconfig_em2="inet 192.168.25.5 netmask 255.255.255.252"
sshd_enable="YES"
# Set dumpdev to "AUTO" to enable crash dumps, "NO" to disable
dumpdev="AUTO"

pf_enable="YES"     # enable pf (load module if required)
pflog_enable="YES"    # start pflogd(8)
gateway_enable="YES"
apache24_enable="YES"
root@firewall:/etc # 
```

- It is already enabled. if not then use following
- Ways to Enable PF

```
sudo kldload pf
sudo pfctl -e
```

-
```
vansil@firewall:~ $ sudo kidload pf
sudo: kidload: command not found
vansil@firewall:~ $ sudo kldload pf
vansil@firewall:~ $ sudo pfctl -e
pf enabled
vansil@firewall:~ $ 
```
- Start pf

```
sudo /etc/rc.d/pf start
```

-
```
vansil@firewall:/etc $ sudo /etc/rc.d/pf start
Enabling pf.
```
- Stop pf

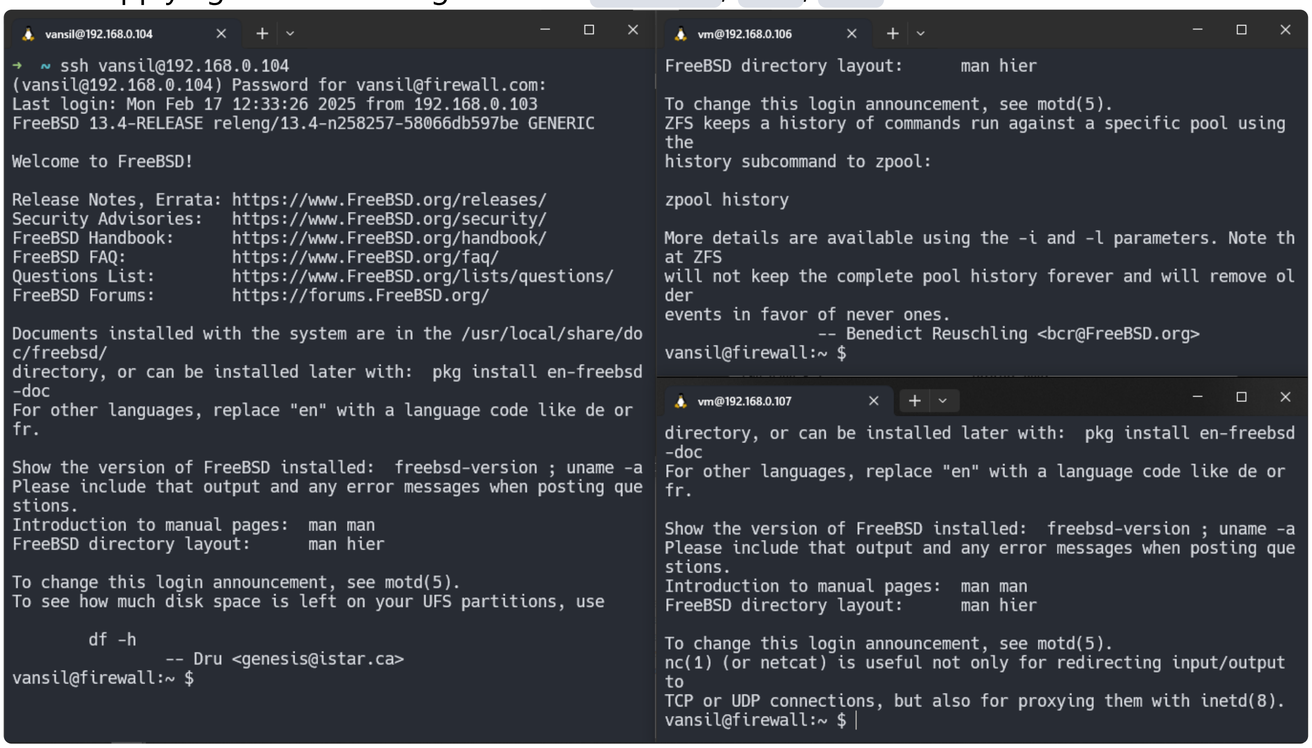
```
sudo /etc/rc.d/pf stop
```

- 

- Edit `rc.conf` to enable `PF`

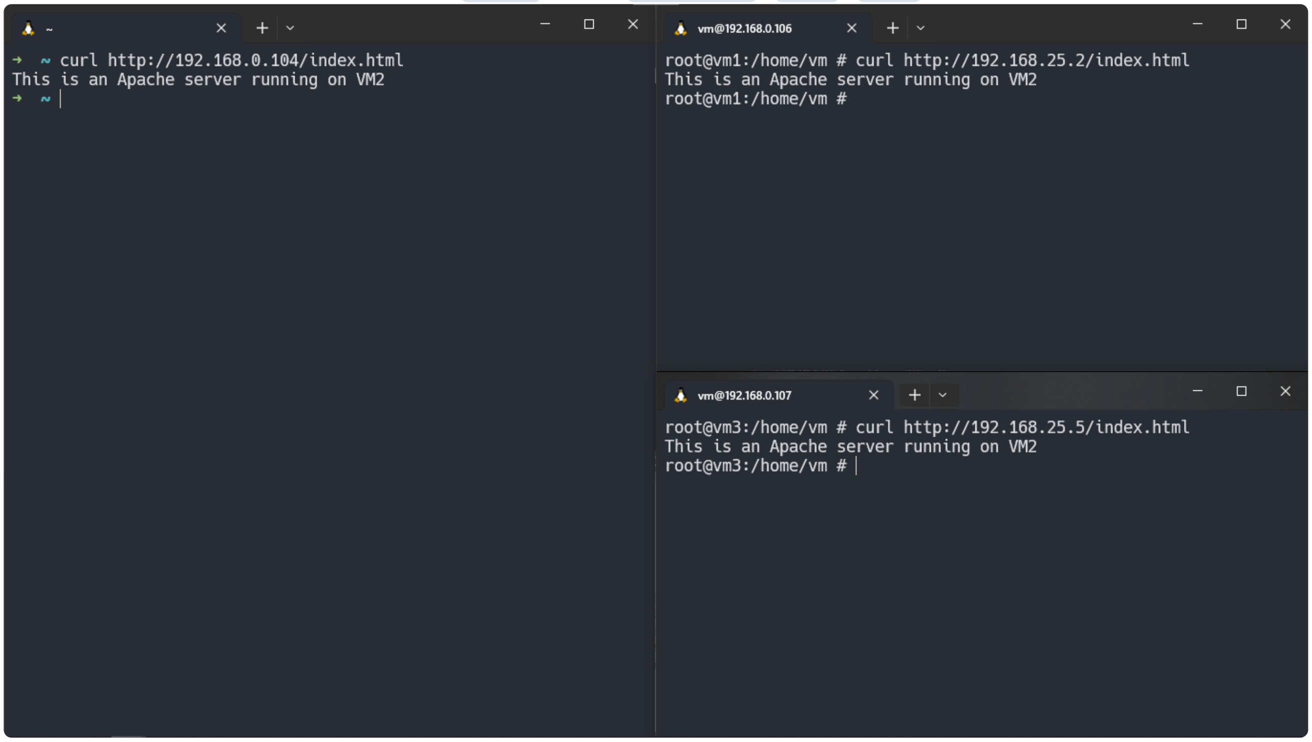
```
sudo sysrc pf_enable="YES"
sudo sysrc pflog_enable="YES"
```

- Configure Rules in `/etc/pf.conf`
- Rules
  - Block all
  - allow ssh 22 only from adapter0
    - from vm1 and vm3 it should be blocked.
  - allow access to port 80
- Before applying firewall... Doing ssh from `Host OS`, `VM1`, `VM3`



- ssh is working!

- Before applying firewall... Doing `curl` from `Host OS`, `VM1`, `VM3`



- `curl` is working for `http` server.
- Let's apply following rules:

```
#pf.conf

block all

# allow http port 80 for internal network
pass in proto tcp from any to any port 80

# allow internal communication (icmp)
pass in proto icmp from any to any
```

- Rules:

```
  GNU nano 8.2                    /etc/pf.conf
block all

pass in proto tcp from any to any port 80

pass in proto icmp from any to any
```

- save and restart pf service.

```
pfctl -nf /etc/pf.conf  # check for syntax errors
pfctl -f /etc/pf.conf
service pf restart
```

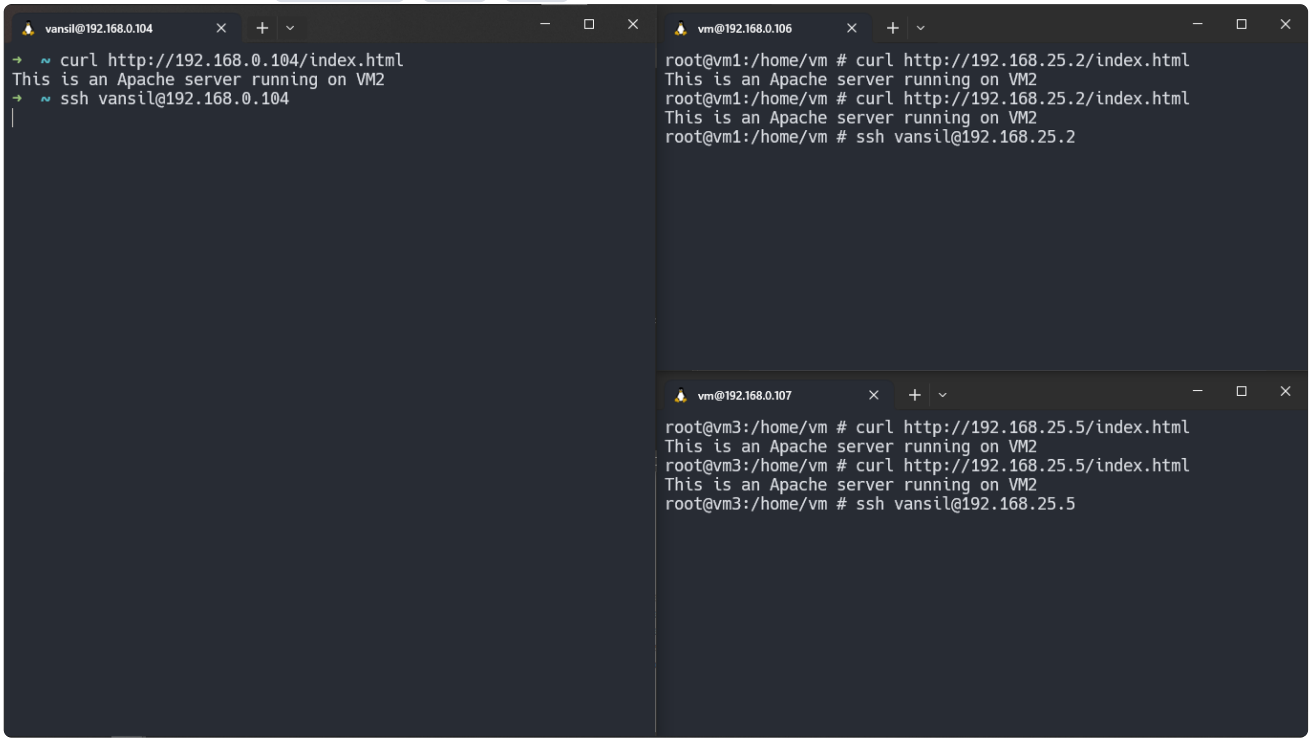
- VM2

```
root@firewall:/home/vansil # cat /etc/pf.conf
block all

pass in on proto tcp from any to any port 80

pass in proto icmp from any to any
root@firewall:/home/vansil # pfctl -f /etc/pf.conf
/etc/pf.conf:3: syntax error
pfctl: Syntax error in config file: pf rules not loaded
root@firewall:/home/vansil # pfctl -nf /etc/pf.conf
/etc/pf.conf:3: syntax error
root@firewall:/home/vansil #
```

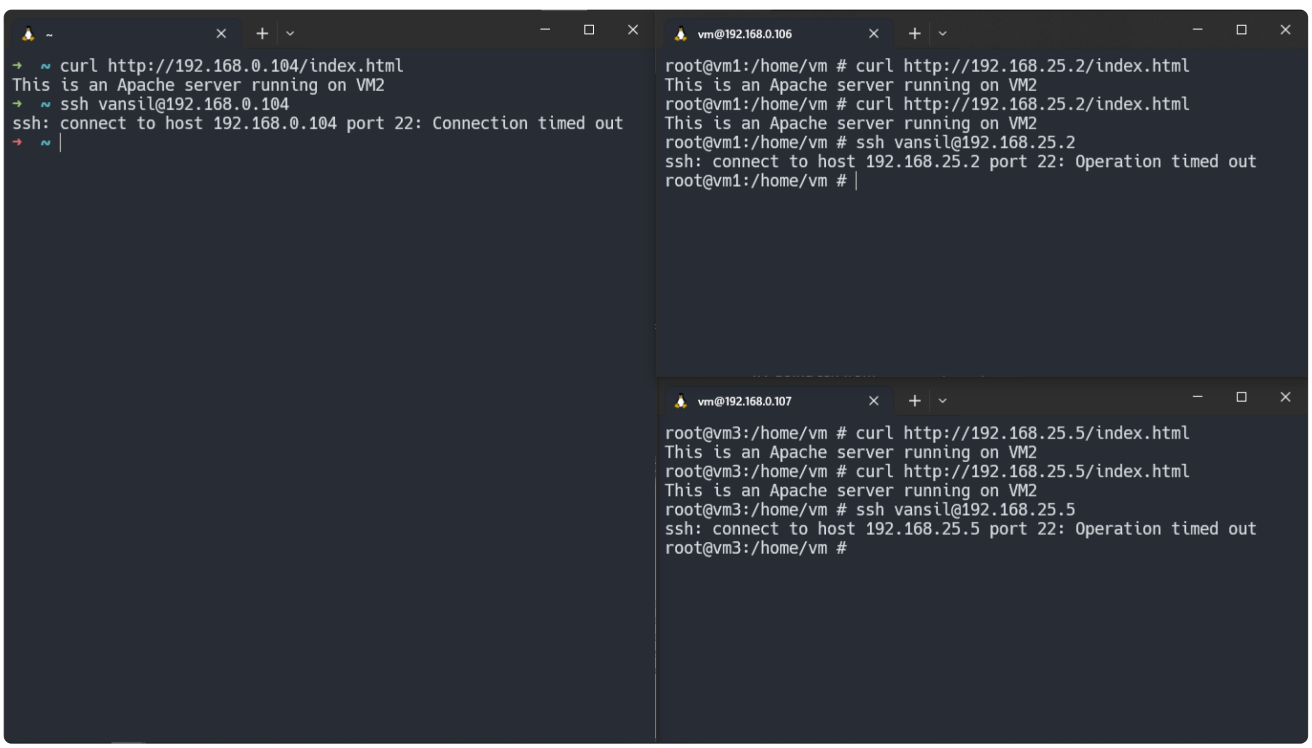
- The moment Applied PF rules... SSH stuck

```
root@firewall:/home/vansil # pfctl -nf /etc/pf.conf
root@firewall:/home/vansil # pfctl -f /etc/pf.conf
```

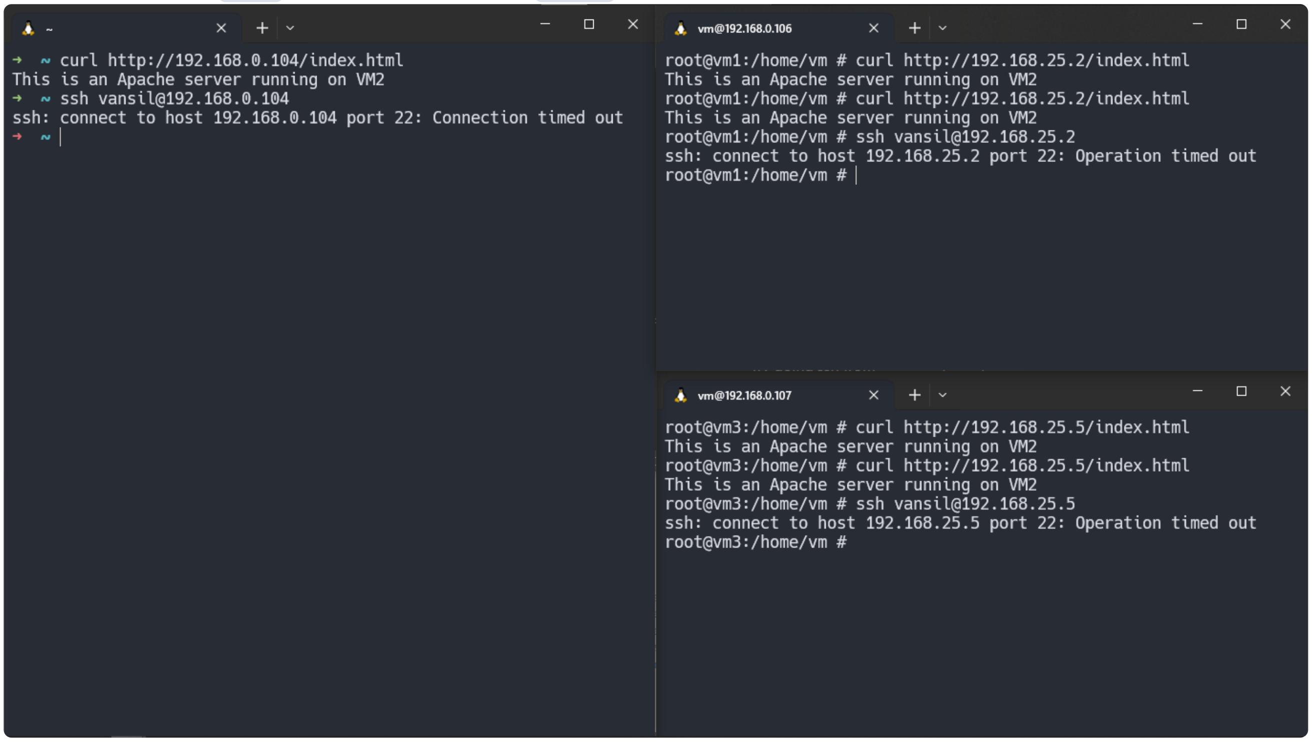
- Restart PF service. `service pf restart`
- Try using `curl` to server using `Main PC`, `VM1`, `VM3`
- Try doing ssh from `Main PC`, `VM1`, `VM3`



- ssh not working from any machines but `http` server is accessible. Says connection timed out.



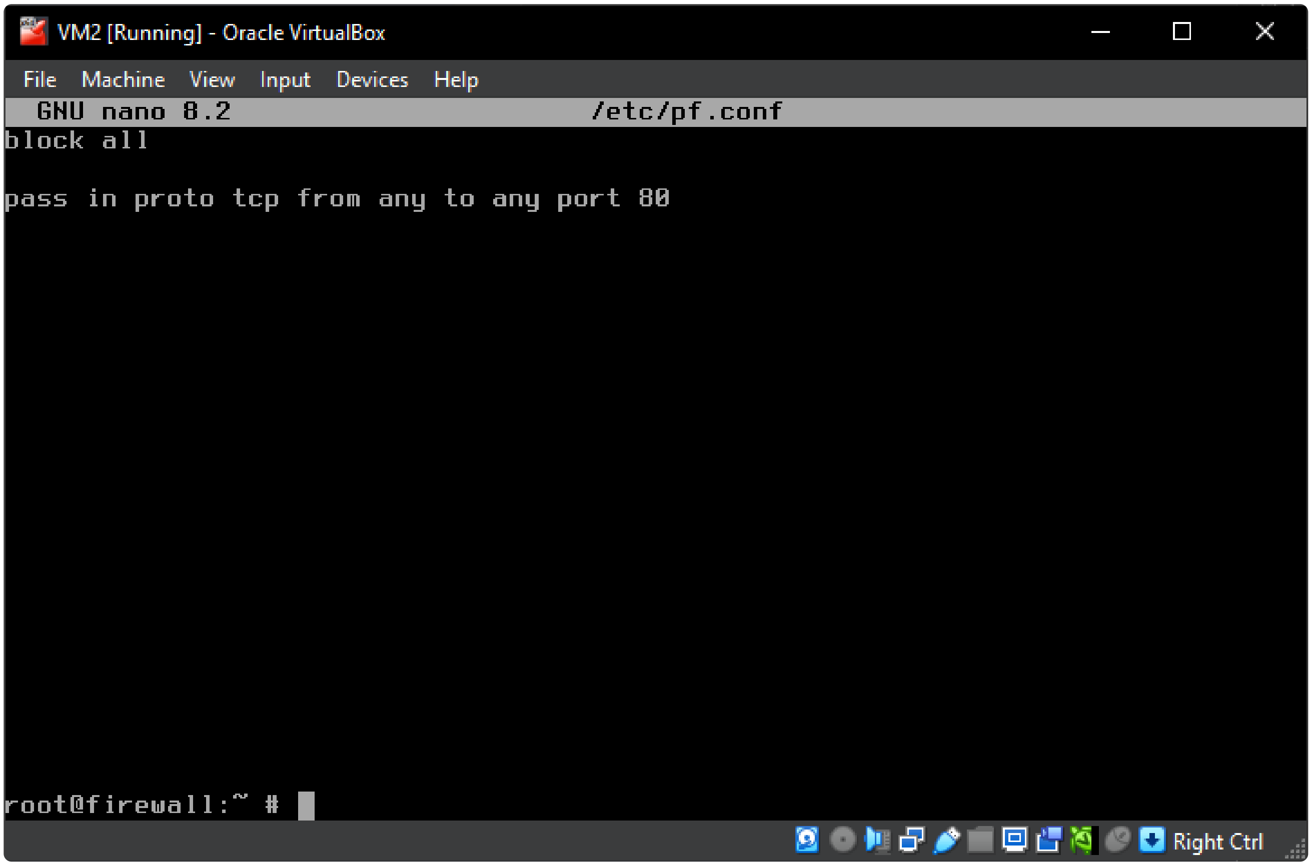
- Do ping work? `yes` because allowed `icmp`



- Disable ping.
- New rules:

```
block all

pass in proto tcp from any to any port 80
```

- VM2

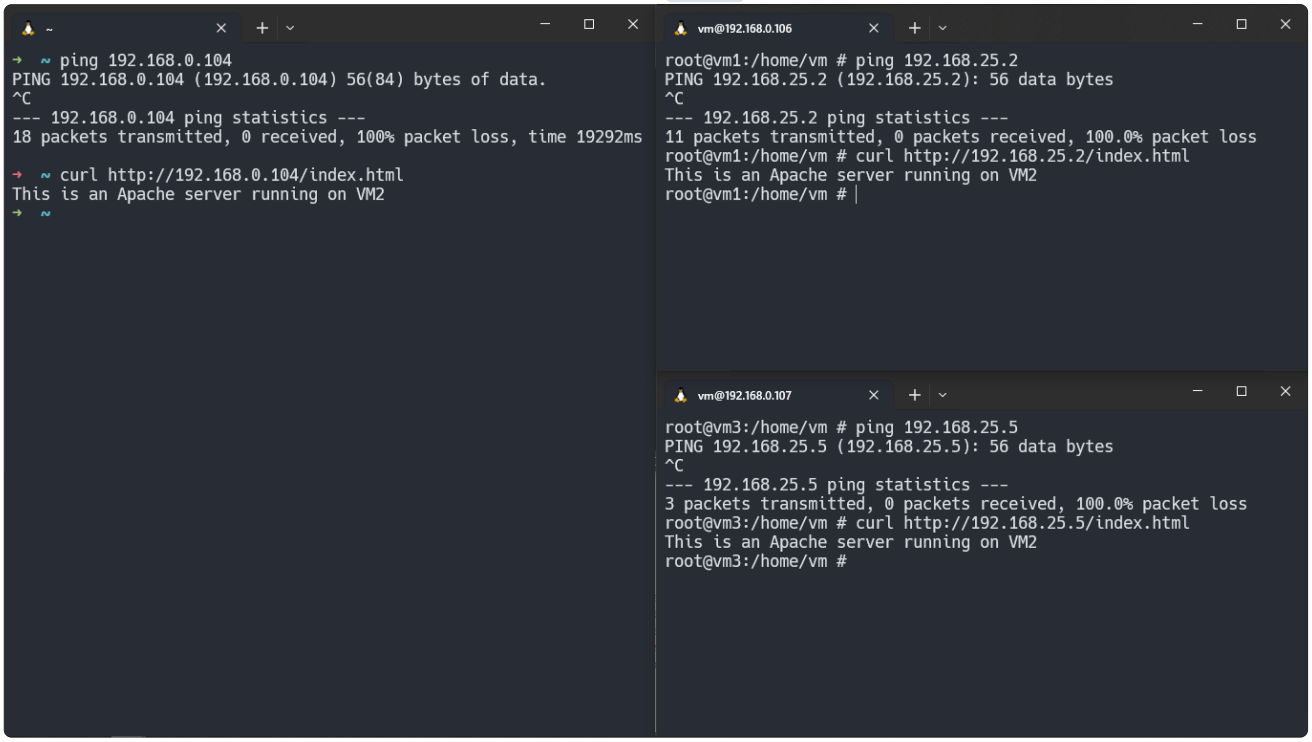


- save and restart pf service.

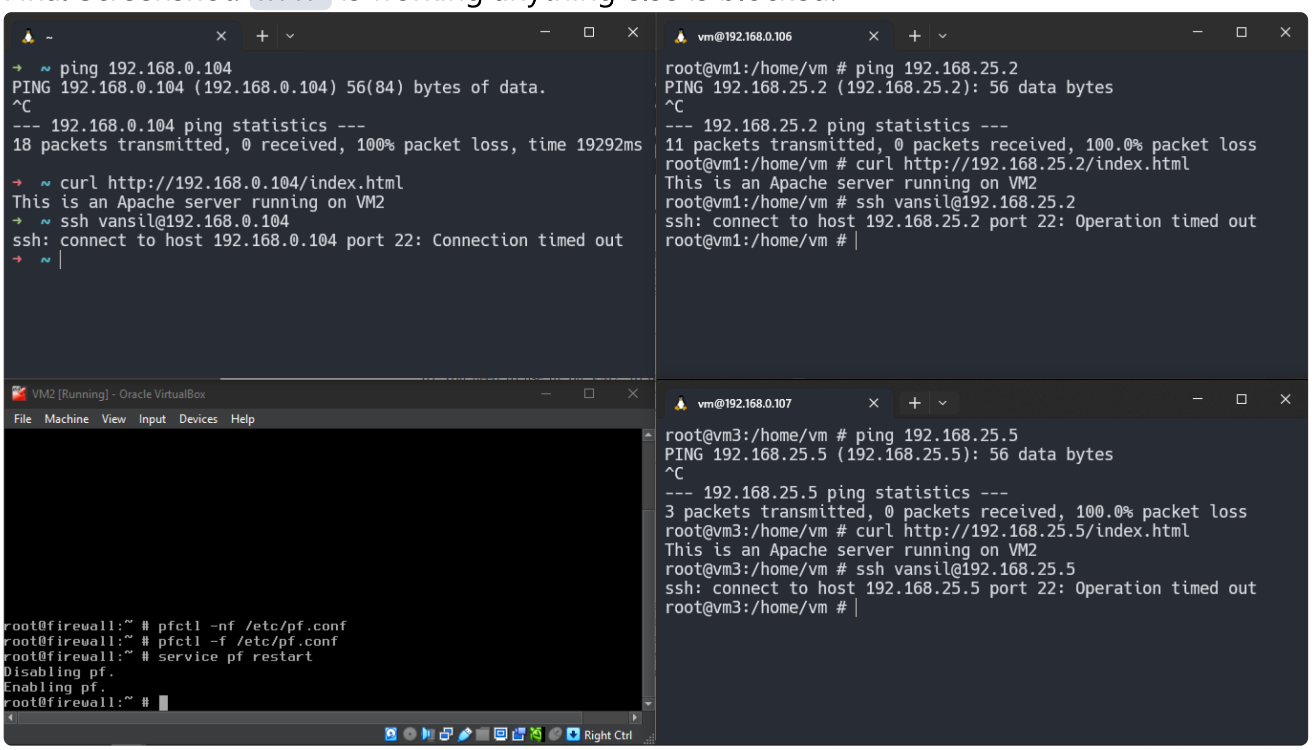
```
pfctl -nf /etc/pf.conf
pfctl -f /etc/pf.conf
service pf restart
```

- VM2

- Check ping? Yes ping is also disabled and only `http` server access is allowed.



- Final Screenshot: `HTTP` is working anything else is blocked.



Artifact Submission: Please create a "professional" report describing the steps involved in achieving each objective, with corresponding screenshots. Importantly, please write the commands used to enable routing (IPv4 forwarding) on VM2 and firewalling rules with a screenshot showing that it did so successfully. You also show a

screenshot where you run a traceroute from VM1 targeted to VM3. (5 Points)

- I have written all the correct steps that led me to completing assignment eventually. But also sharing extra pdf 24m0847 CS790 Assignment 1 Learnings to verify my attempts and learning in coming to approximately correct steps.
